

Introduction to DIP

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From Gonzales, Woods & web



FIGURE 1.1 A digital picture produced in 1921 from a coded tape by a telegraph printer with special type faces. (McFarlane.[†])

FIGURE 1.2 A digital picture made in 1922 from a tape punched after the signals had crossed the Atlantic twice. (McFarlane.)



FIGURE 1.3

Unretouched
cable picture of
Generals Pershing
and Foch,
transmitted in
1929 from
London to New
York by 15-tone
equipment.
(McFarlane.)



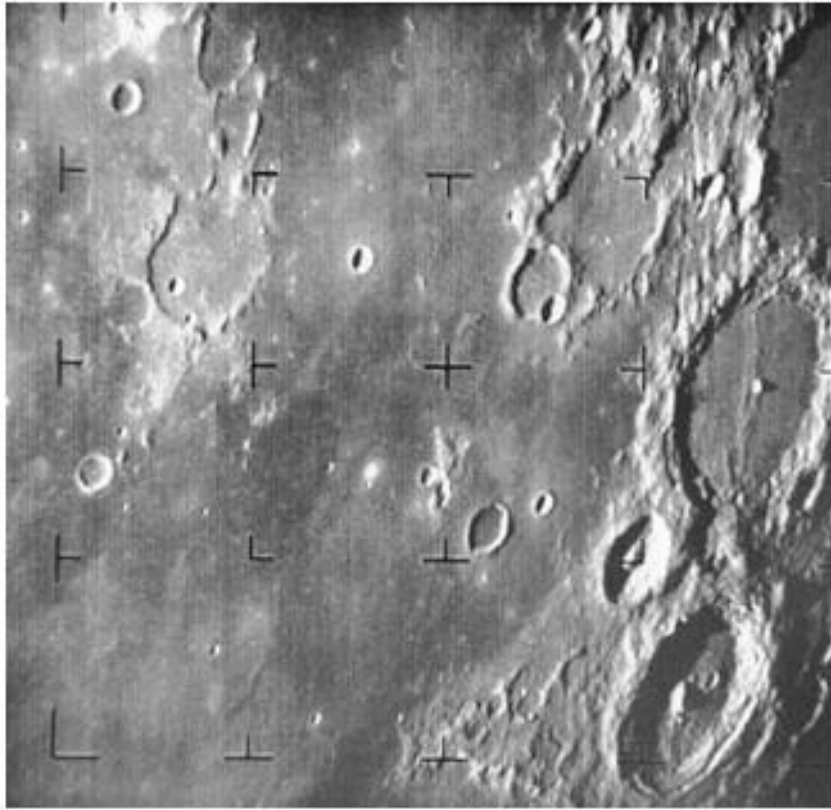


FIGURE 1.4 The first picture of the moon by a U.S. spacecraft. *Ranger 7* took this image on July 31, 1964 at 9:09 A.M. EDT, about 17 minutes before impacting the lunar surface. (Courtesy of NASA.)

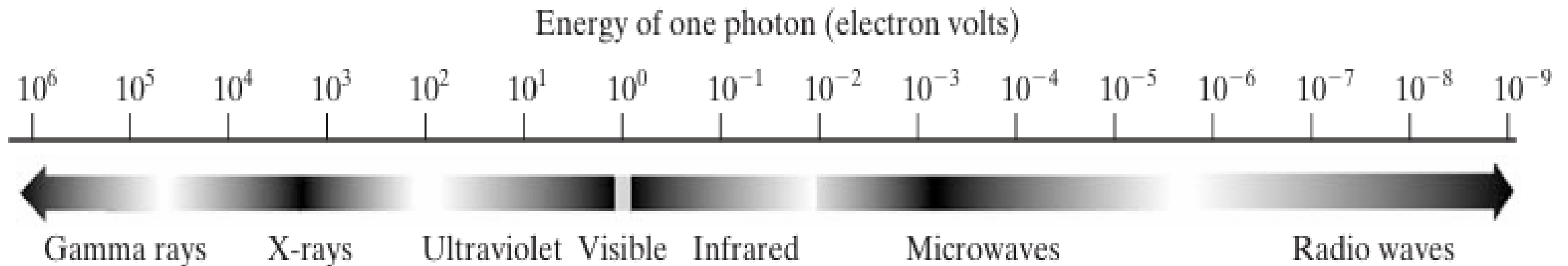
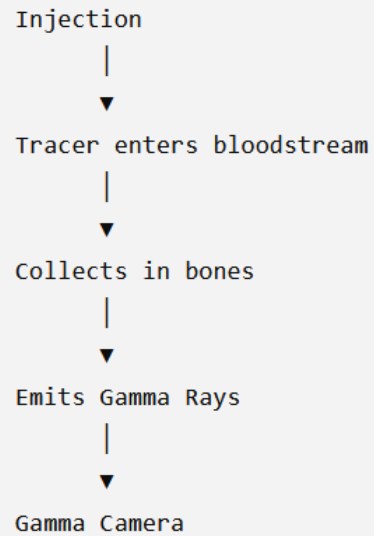


FIGURE 1.5 The electromagnetic spectrum arranged according to energy per photon.

(a) Bone Scan

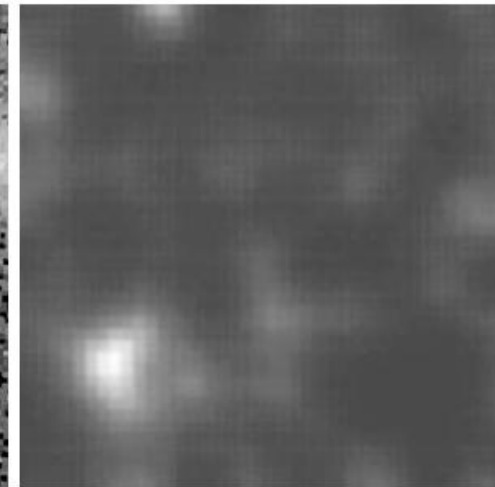
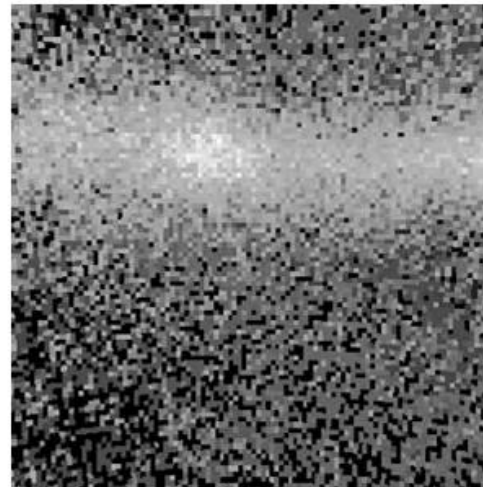
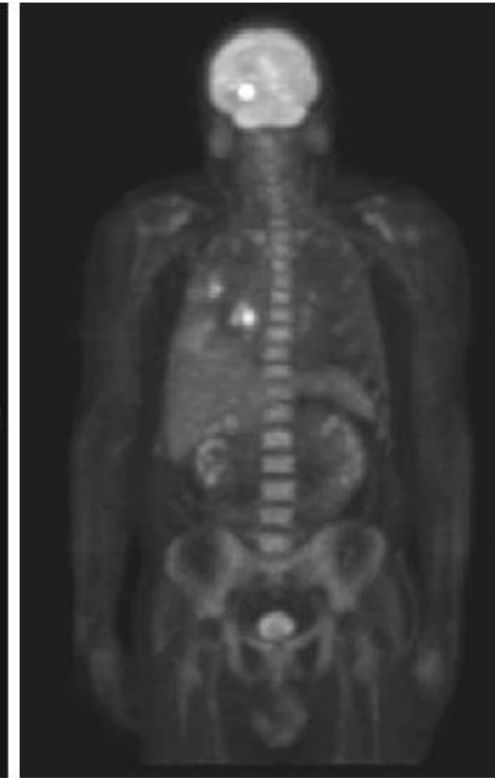
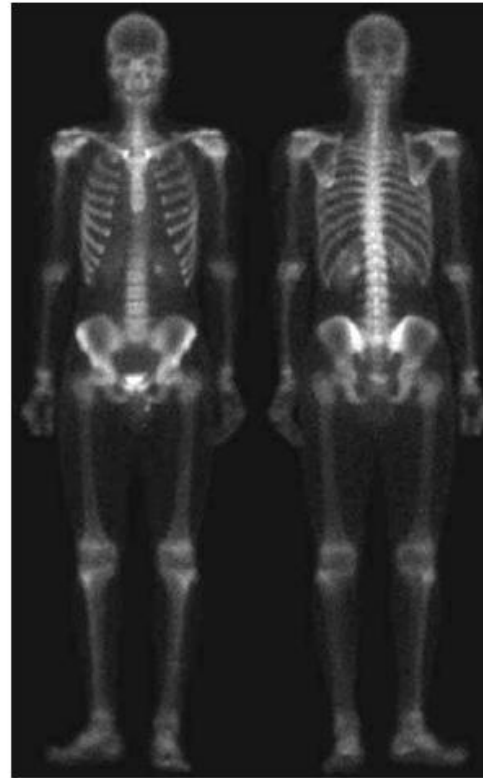
The patient is injected with a **radioactive tracer**



a b
c d

FIGURE 1.6

Examples of gamma-ray imaging. (a) Bone scan. (b) PET image. (c) Cygnus Loop. (d) Gamma radiation (bright spot) from a reactor valve. (Images courtesy of (a) G.E. Medical Systems, (b) Dr. Michael E. Casey, CTI PET Systems, (c) NASA, (d) Professors Zhong He and David K. Wehe, University of Michigan.)



(b) PET Image

PET = Positron Emission Tomography

Instead of bones,

the tracer accumulates in **metabolically active tissues**.

Radioactive Glucose



Cancer cells consume more glucose



More Gamma Rays



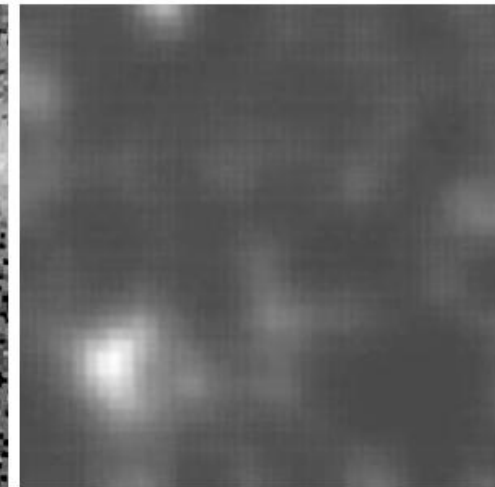
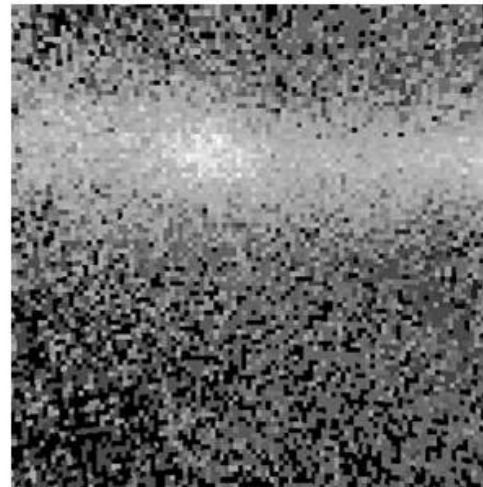
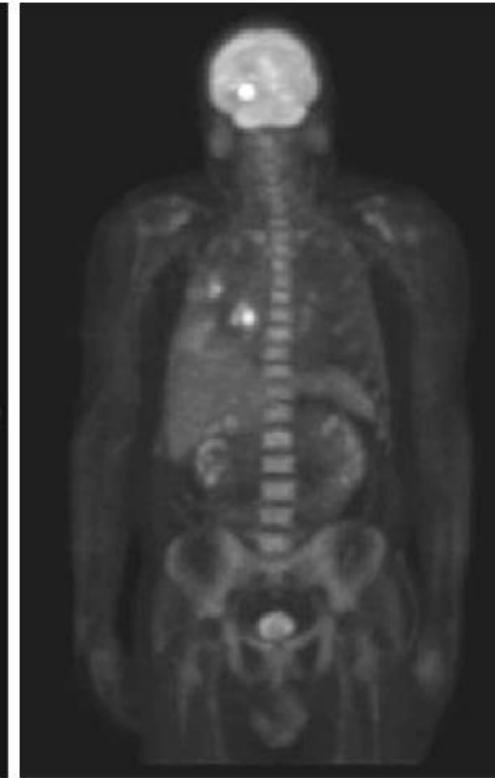
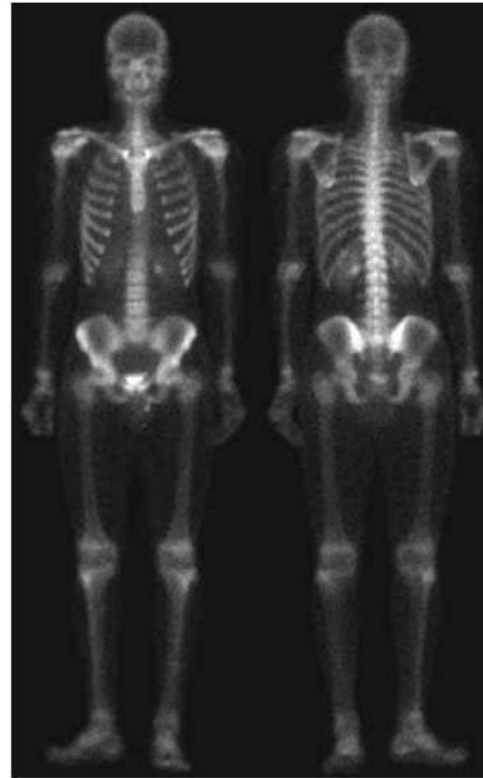
Bright Spot



a b
c d

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(c) Cygnus Loop

This is **not** a medical image.

It is an astronomical gamma-ray image.

Space



Gamma Rays



NASA Detector



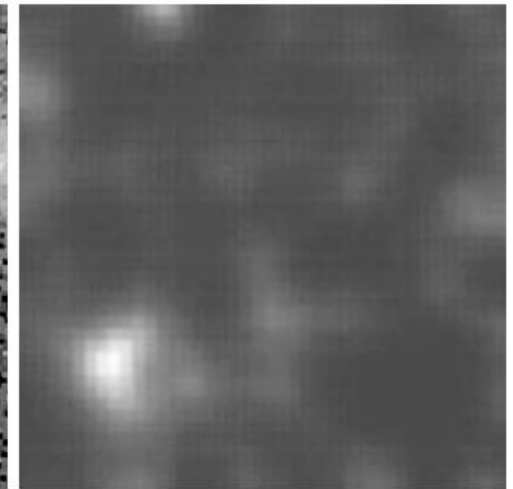
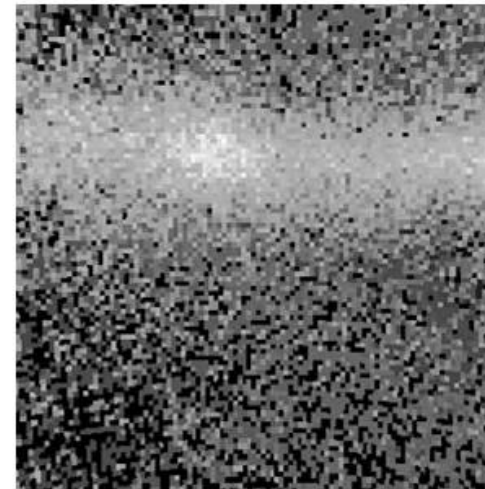
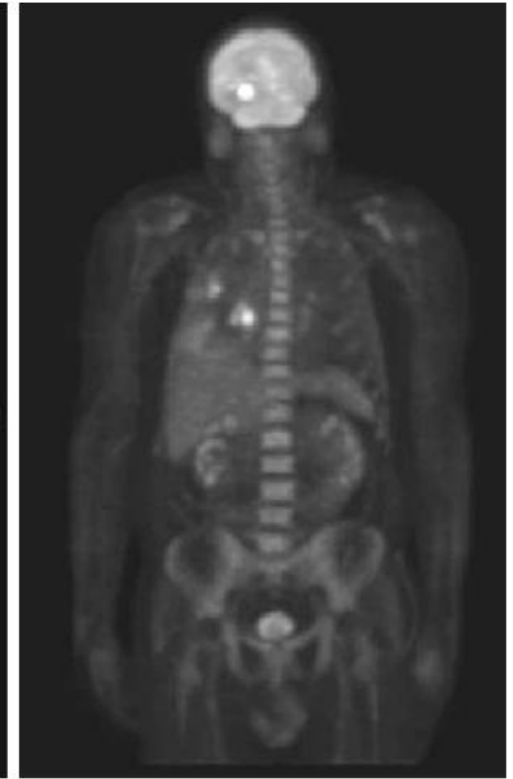
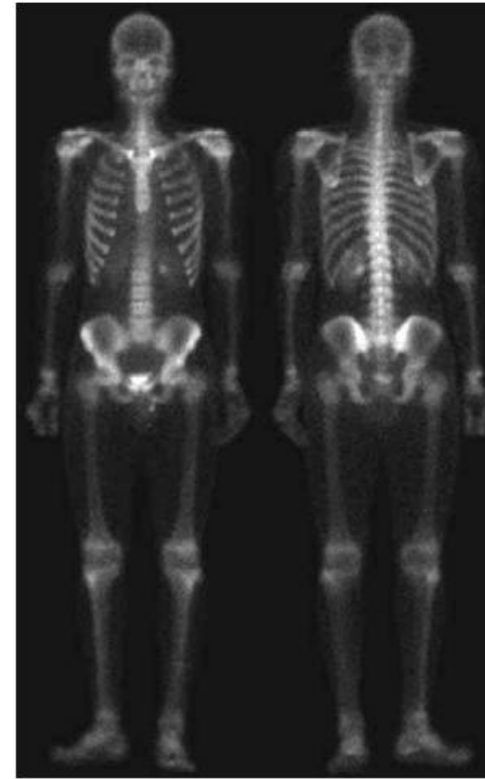
Gamma Image

The Cygnus Loop is a **supernova remnant**—expanding remains of a massive star that exploded.

a b
c d

FIGURE 1.6

Examples of gamma-ray imaging. (a) Bone scan. (b) PET scan. (c) Cygnus Loop. (d) Gamma radiation (bright spot) from a reactor valve. Images courtesy of (a) G.E. Medical Systems, (b) Dr. Michael L. Casey, CTI PET Systems, (c) NASA, (d) Professors Zhong He and David K. Wehe, University of Michigan.)



(d) Reactor Valve

This is an industrial application.

Nuclear Reactor Valve



Gamma Radiation Leak



Gamma Camera

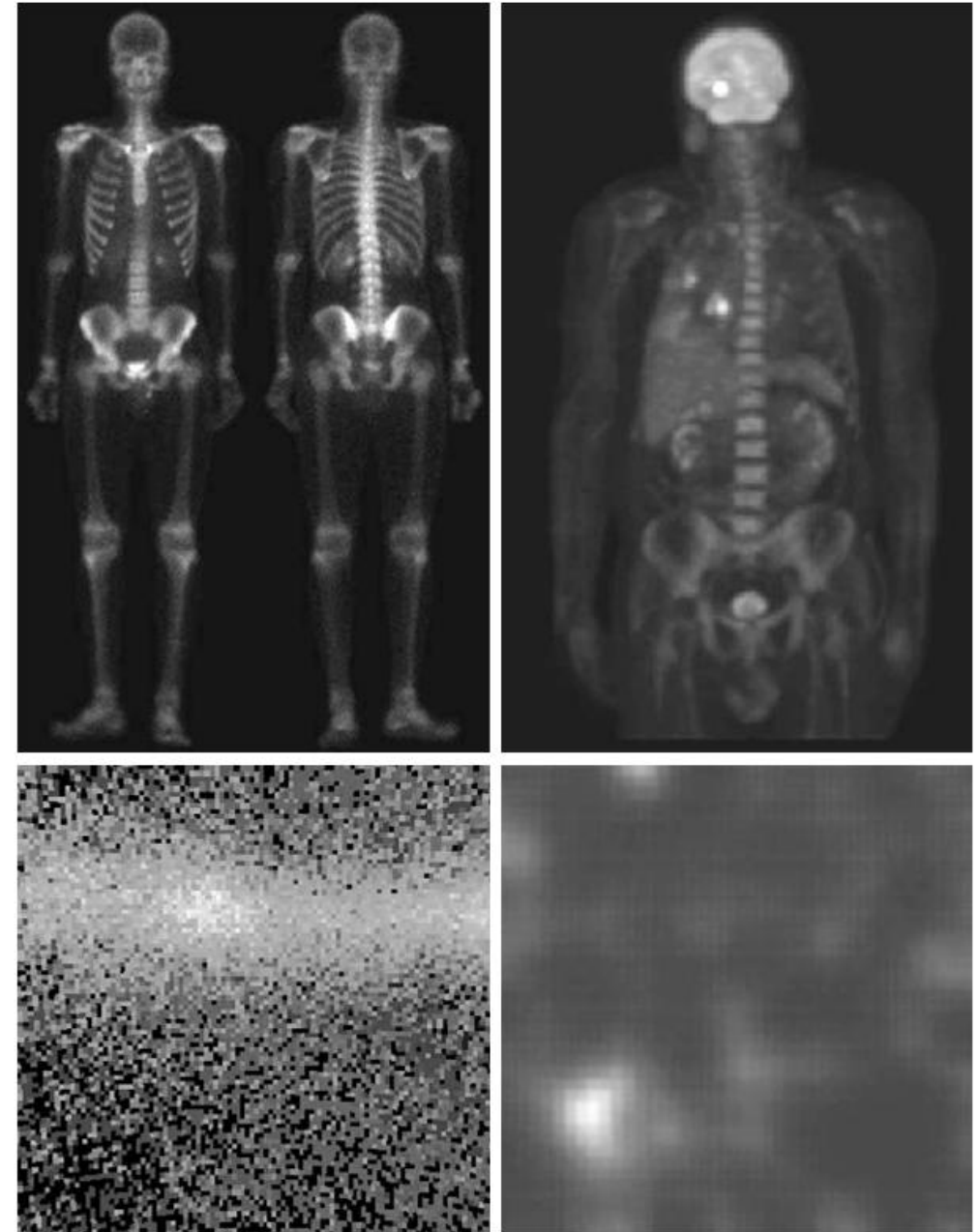


Bright Spot

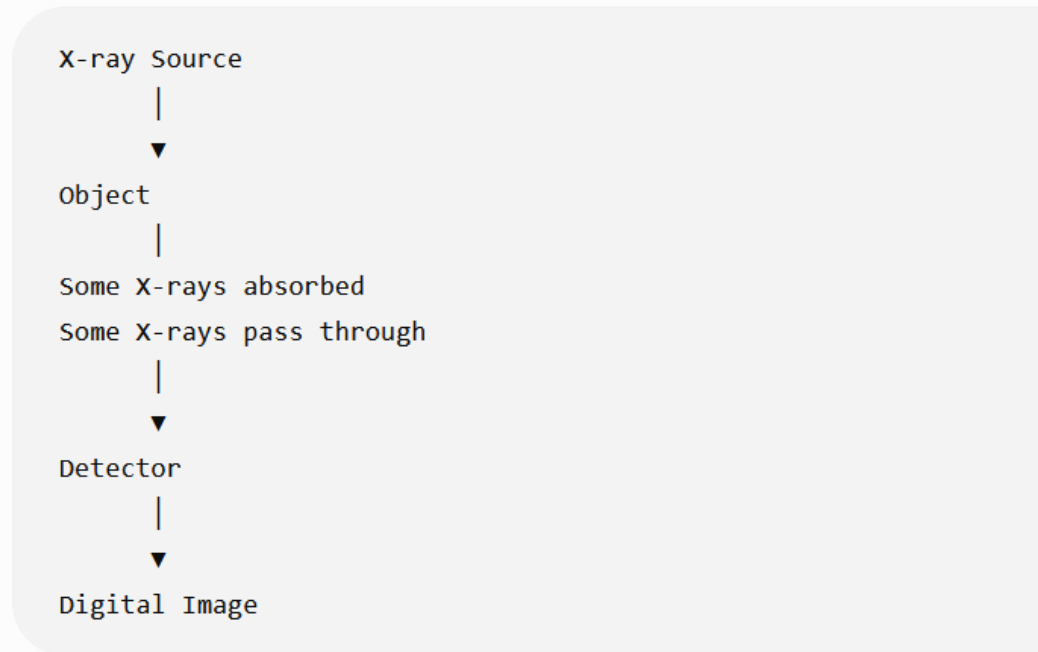
a b
c d

FIGURE 1.6

Examples of gamma-ray imaging. (a) Bone scan. (b) PET scan. (c) Cygnus Loop. (d) Gamma radiation (bright spot) from a reactor valve. (Images courtesy of (a) G.E. Medical Systems, (b) Dr. Michael E. Casey, CTI PET Systems, (c) NASA, (d) Professors Zhong He and David K. Wehe, University of Michigan.)



Principle of X-ray Imaging



Dense materials absorb more X-rays → appear **white**.

Less dense materials allow more X-rays to pass → appear **dark**.

a) Medical Imaging, b)
Aortic Angiogram-Injecting
Contrast dye, c) Braine CT
Scan (without or with
contrast)

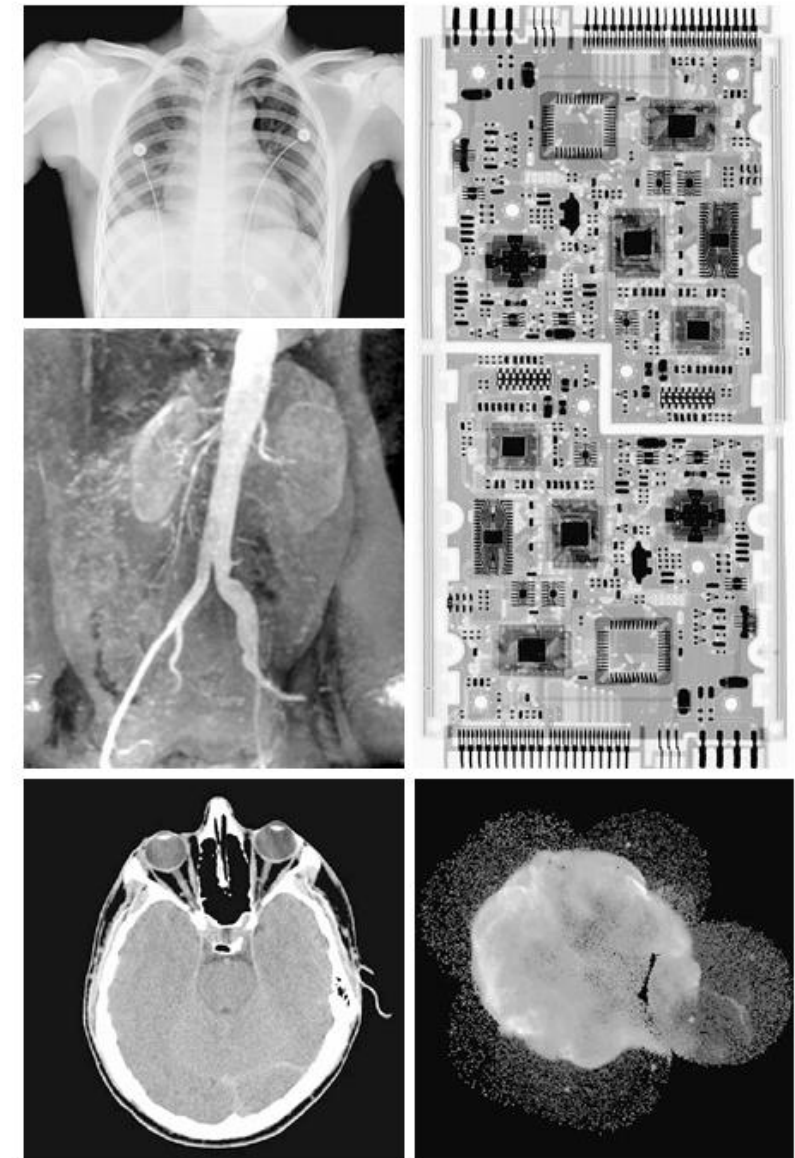
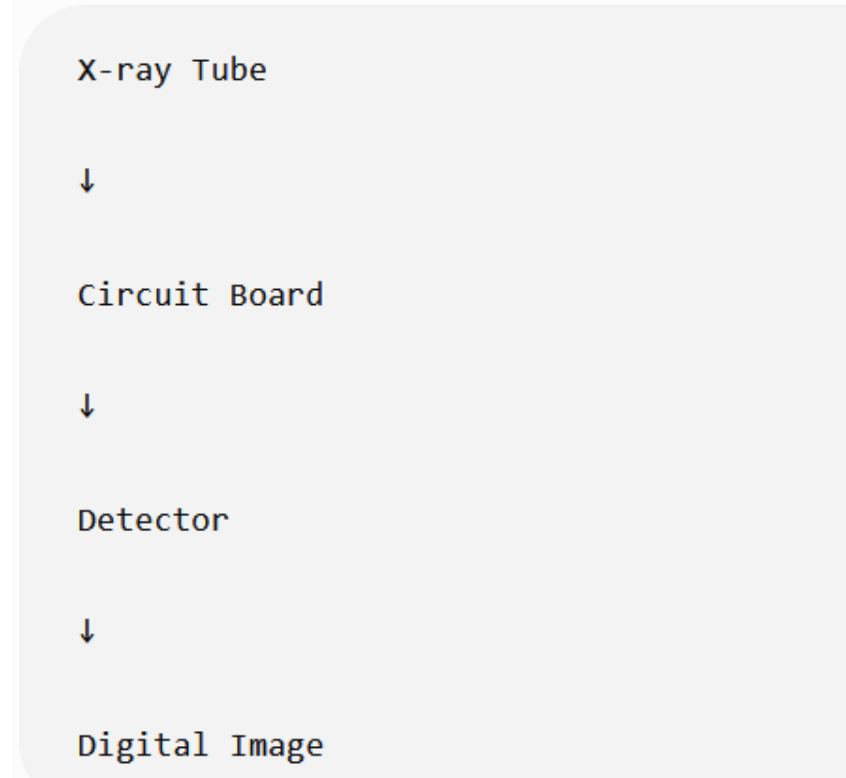


FIGURE 1.7 Examples of X-ray imaging. (a) Chest X-ray. (b) Aortic angiogram. (c) Head CT. (d) Circuit boards. (e) Cygnus Loop. (Images courtesy of (a) and (c) Dr. David R. Pickens, Dept. of Radiology & Radiological Sciences, Vanderbilt University Medical Center; (b) Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School; (d) Mr. Joseph E. Pascente, Lixi, Inc.; and (e) NASA.)

d) X Ray- PCB- Industrial Application

Exactly the same principle as a chest X-ray.



e) Astronomical X Ray- through space X ray Telescope

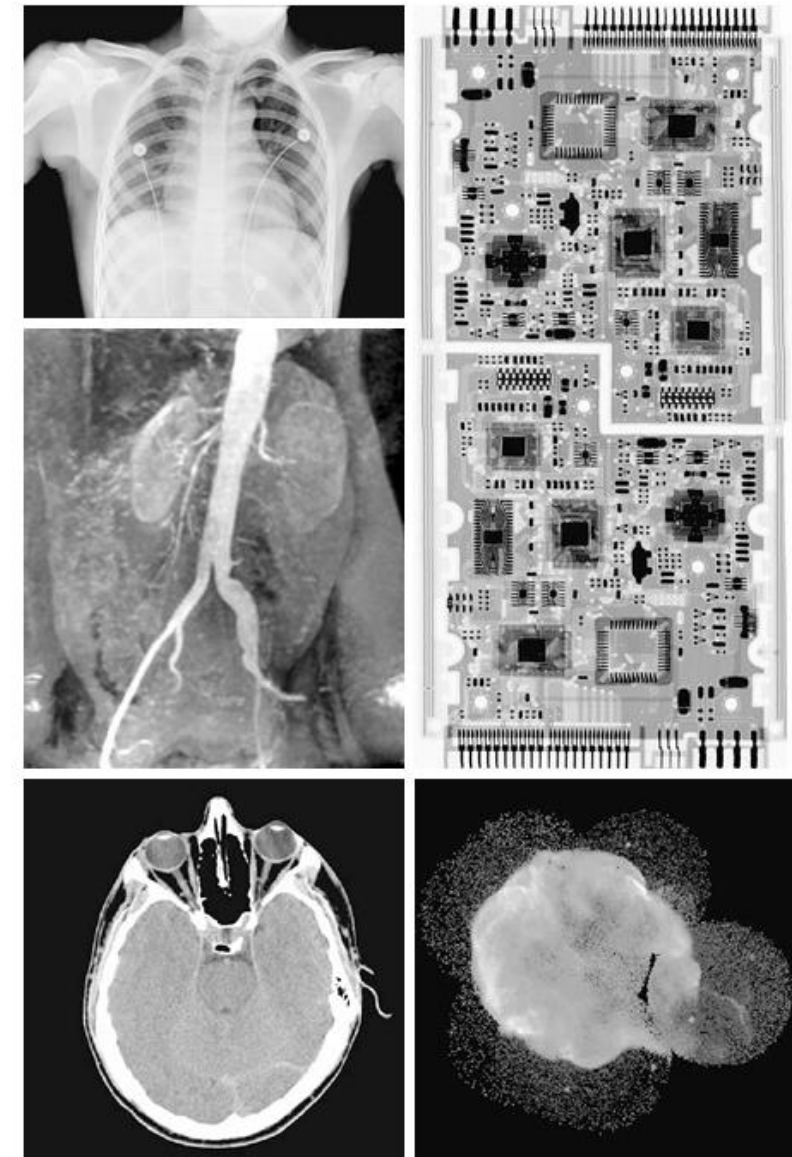


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Principle of UV Imaging

UV Source (Sun or UV Lamp)



Object



Reflects / Emits UV



UV Camera



Digital Image

(b) Smut Corn

"Smut" is a fungal disease caused by **corn smut** (*Ustilago maydis*).

Fungus attacks corn



Surface chemistry changes



Different UV reflection



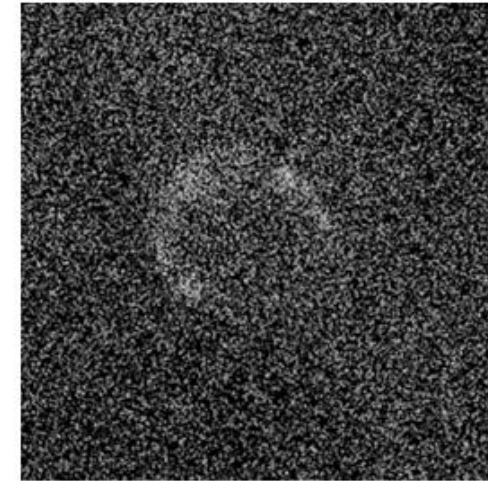
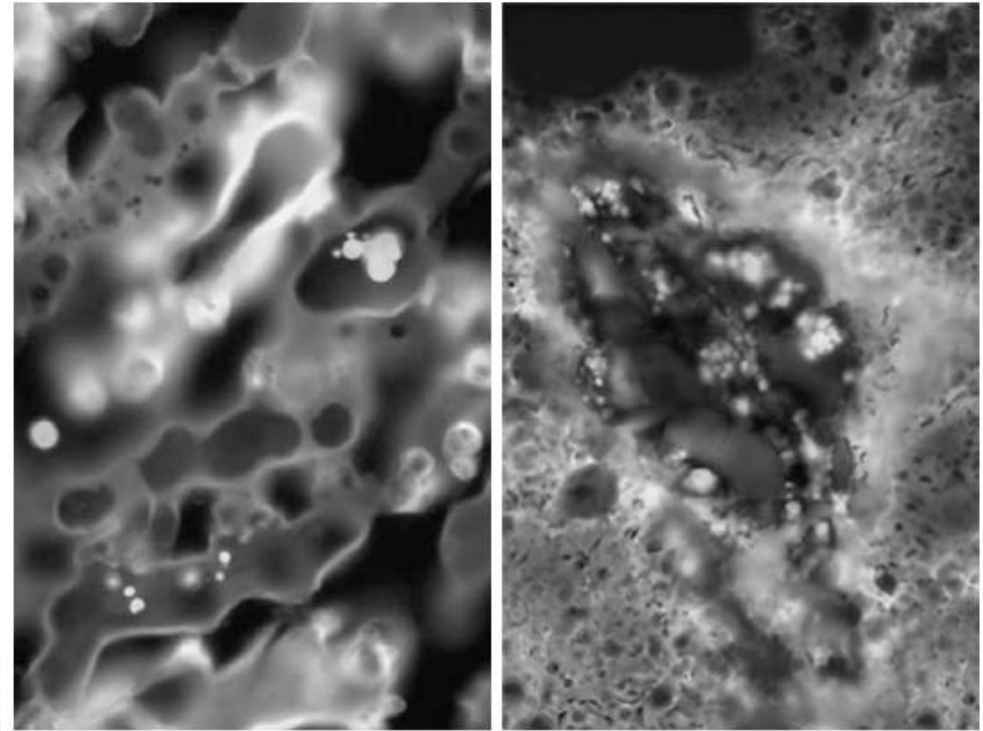
Disease becomes visible

a b
c

FIGURE 1.8

Examples of ultraviolet imaging.

(a) Normal corn.
(b) Smut corn.
(c) Cygnus Loop.
(Images courtesy of (a) and (b) Dr. Michael W. Davidson, Florida State University, (c) NASA.)



grasses, onions, and sorghum that can be caused by any of more than 700 species

Principle of Light Microscopy

Light Source



Small Object



Light reflected / transmitted



Objective Lens



Eyepiece / Camera



Digital Image

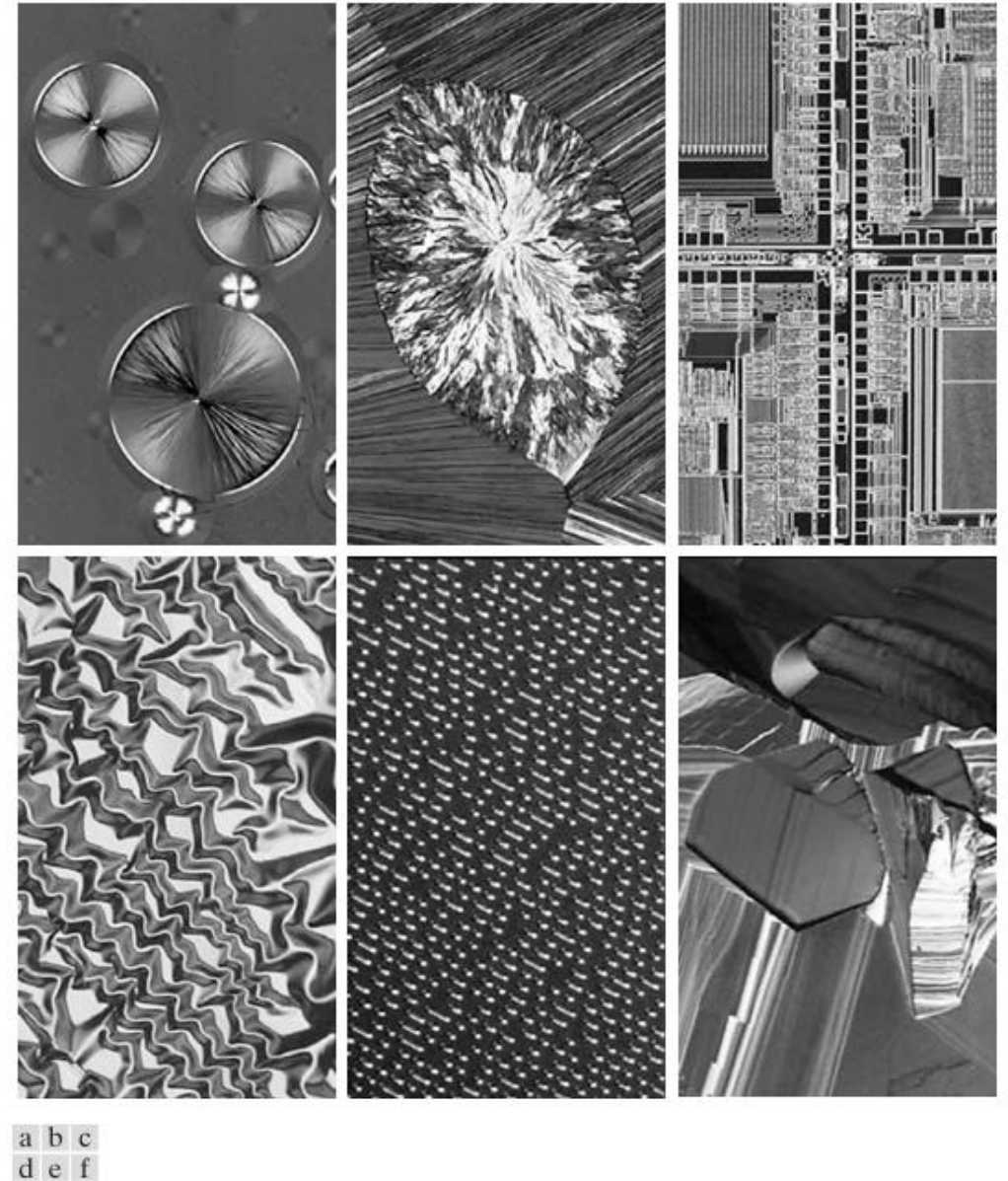


FIGURE 1.9 Examples of light microscopy images. (a) Taxol (anticancer agent), magnified 250 $\times$. (b) Cholesterol—40 $\times$. (c) Microprocessor—60 $\times$. (d) Nickel oxide thin film—600 $\times$. (e) Surface of audio CD—1750 $\times$. (f) Organic superconductor—450 $\times$. (Images courtesy of Dr. Michael W. Davidson, Florida State University.)

Figure 1.9 – Examples of Light Microscopy Images

Fig.	Sample	Magnification	What does it show?	Why is it important?	Applications
(a)	Taxol (anticancer drug)	250×	Crystal structure of Taxol drug particles	Study crystal growth, purity, and morphology	Pharmaceutical manufacturing, drug quality control
(b)	Cholesterol Crystal	40×	Shape and arrangement of cholesterol crystals	Analyze crystal morphology and medical deposits	Medical research, pathology, cholesterol studies
(c)	Microprocessor	60×	Integrated circuit (IC) layout showing transistors, metal tracks and connections	Inspect chip fabrication and detect manufacturing defects	Semiconductor industry, VLSI inspection, quality control
(d)	Nickel Oxide Thin Film	600×	Surface morphology and grain structure of a thin material layer	Study material properties and defects	Materials science, batteries, sensors, solar cells
(e)	Surface of Audio CD	1750×	Microscopic pits and lands storing digital information	Verify data-storage patterns and manufacturing quality	Optical storage (CD/DVD), data storage technology
(f)	Organic Superconductor	450×	Crystal morphology and surface structure of superconducting material	Understand crystal growth and material characteristics	Superconductivity research, quantum materials, advanced electronics

TABLE 1.1

Thematic bands
in NASA's
LANDSAT
satellite.

Band No.	Name	Wavelength (μm)	Characteristics and Uses
1	Visible blue	0.45–0.52	Maximum water penetration
2	Visible green	0.52–0.60	Good for measuring plant vigor
3	Visible red	0.63–0.69	Vegetation discrimination
4	Near infrared	0.76–0.90	Biomass and shoreline mapping
5	Middle infrared	1.55–1.75	Moisture content of soil and vegetation
6	Thermal infrared	10.4–12.5	Soil moisture; thermal mapping
7	Middle infrared	2.08–2.35	Mineral mapping

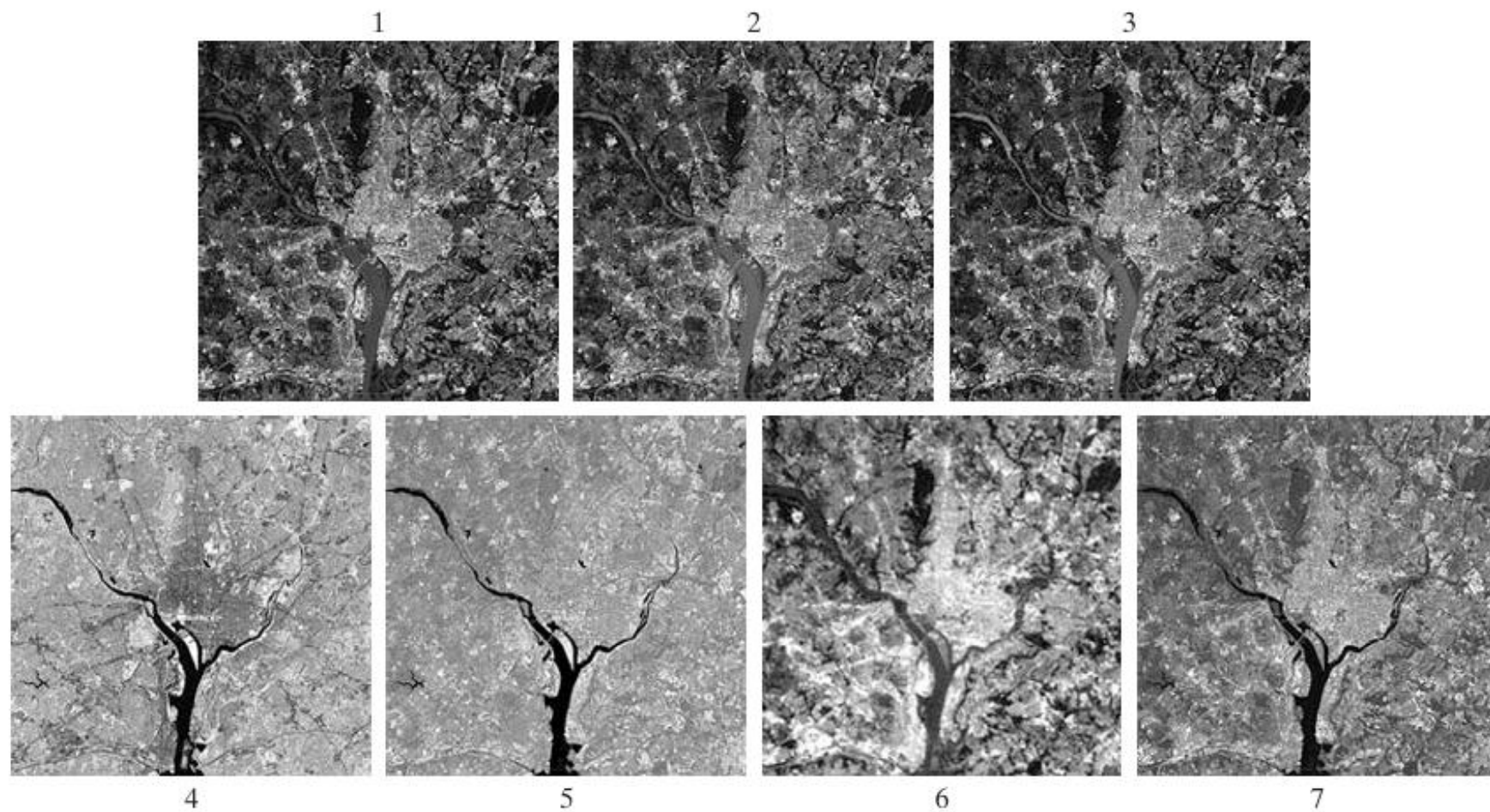


FIGURE 1.10 LANDSAT satellite images of the Washington, D.C. area. The numbers refer to the thematic bands in Table 1.1. (Images courtesy of NASA.)

1.3 ■ Examples of Fields that Use Digital Image Processing 15



FIGURE 1.11
Satellite image
of Hurricane
Katrina taken on
August 29, 2005.
(Courtesy of
NOAA.)

FIGURE 1.12
Infrared satellite
images of the
Americas. The
small gray map is
provided for
reference.
(Courtesy of
NOAA.)

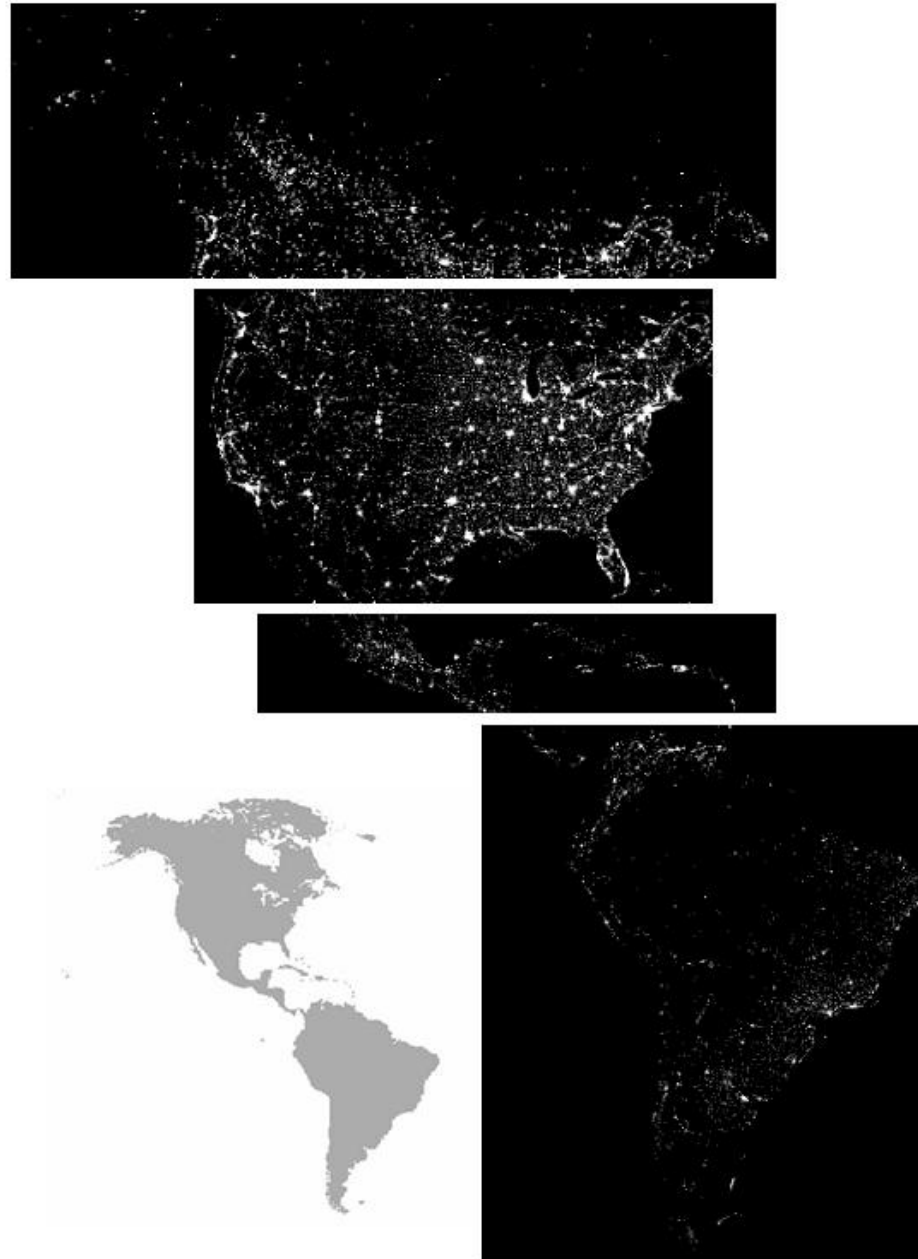


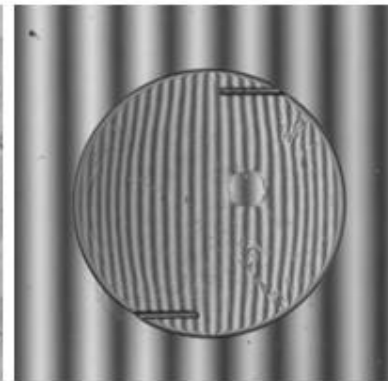
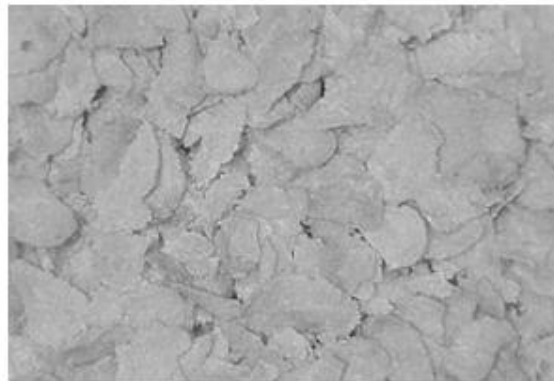
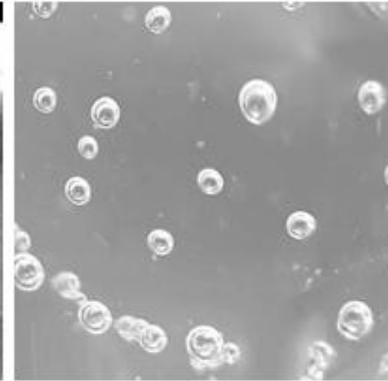
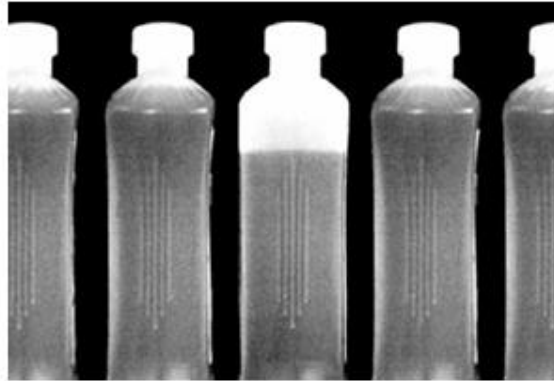
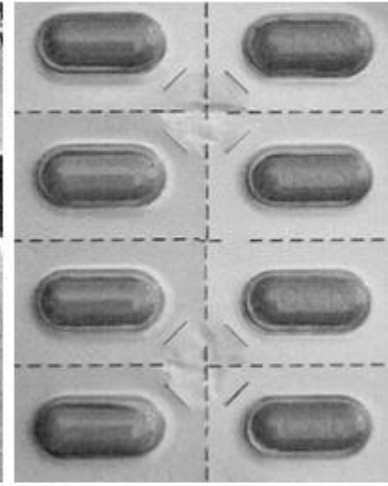
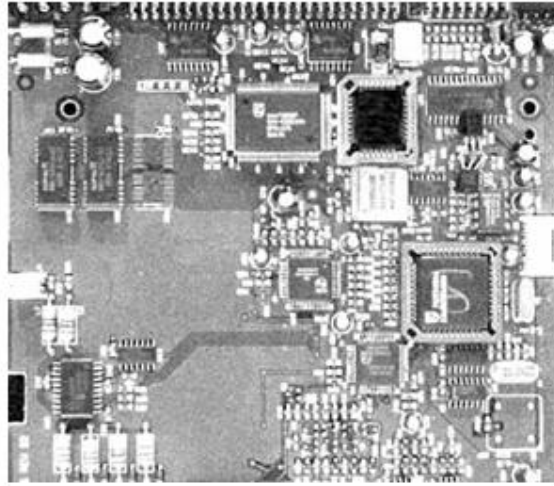


FIGURE 1.13

Infrared satellite images of the remaining populated part of the world. The small gray map is provided for reference. (Courtesy of NOAA.)

a	b
c	d
e	f

FIGURE 1.14
Some examples
of manufactured
goods often
checked using
digital image
processing.
(a) A circuit
board controller.
(b) Packaged pills.
(c) Bottles.
(d) Air bubbles
in a clear-plastic
product.
(e) Cereal.
(f) Image of
intraocular
implant.
(Fig. (f) courtesy
of Mr. Pete Sites,
Perceptics
Corporation.)





a b
c
d

FIGURE 1.15

Some additional examples of imaging in the visual spectrum.

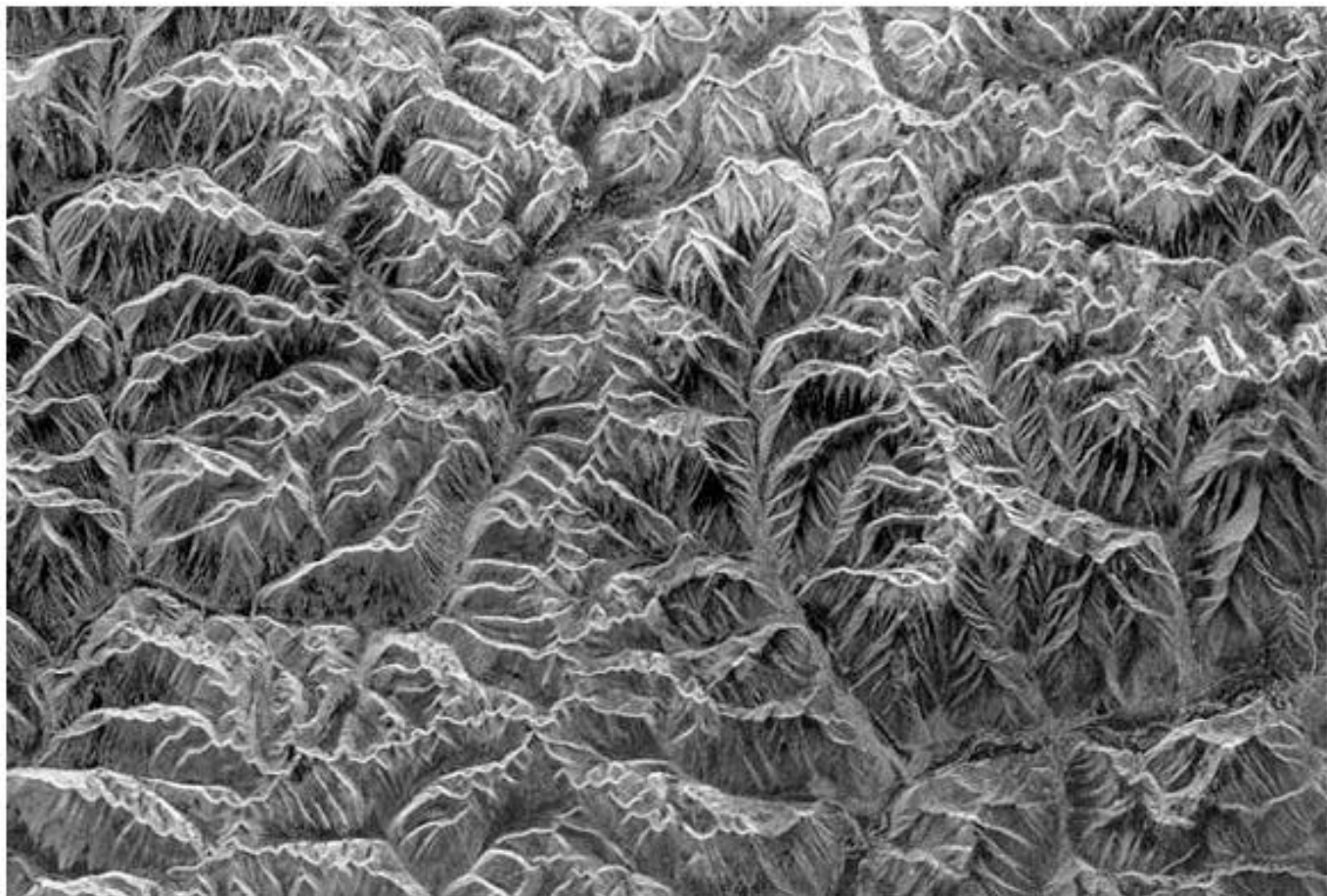
(a) Thumb print. (b) Paper currency. (c) and (d) Automated license plate reading.

(Figure (a) courtesy of the National Institute of Standards and Technology.

Figures (c) and (d) courtesy of Dr. Juan Herrera, Perceptics Corporation.)



FIGURE 1.16
Spaceborne radar
image of
mountains in
southeast Tibet.
(Courtesy of
NASA.)





a b

FIGURE 1.17 MRI images of a human (a) knee, and (b) spine. (Image (a) courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School, and (b) courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

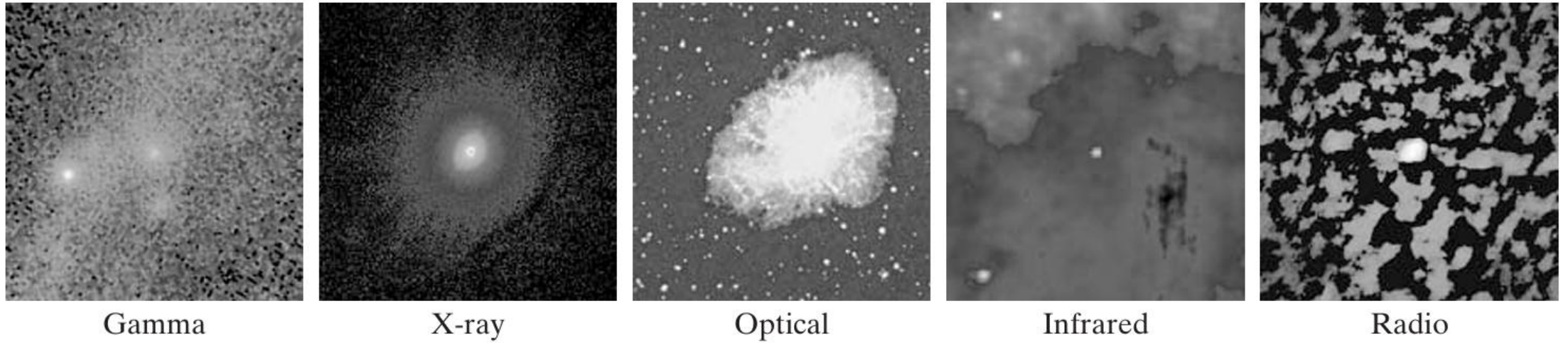
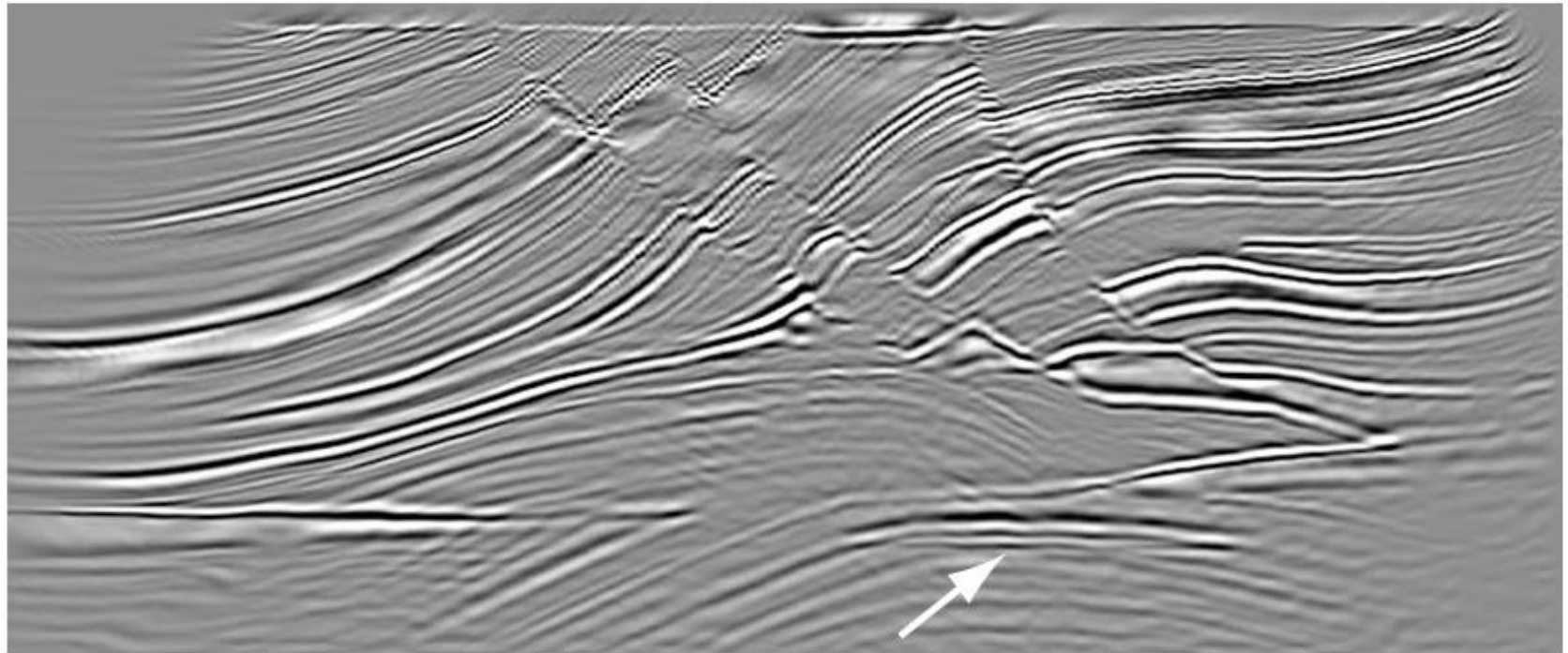
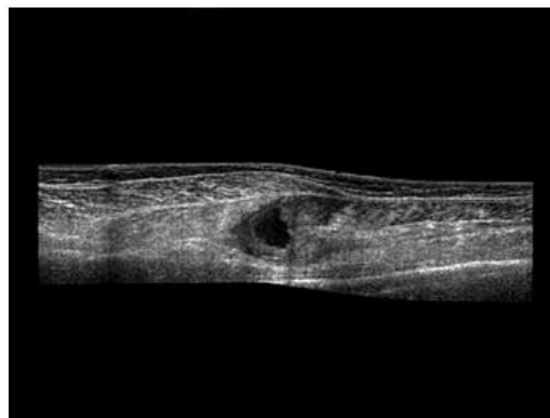
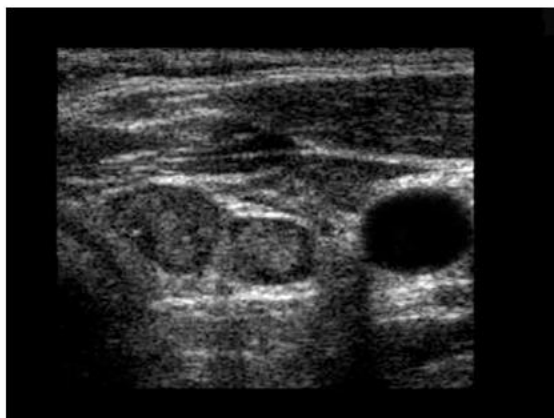
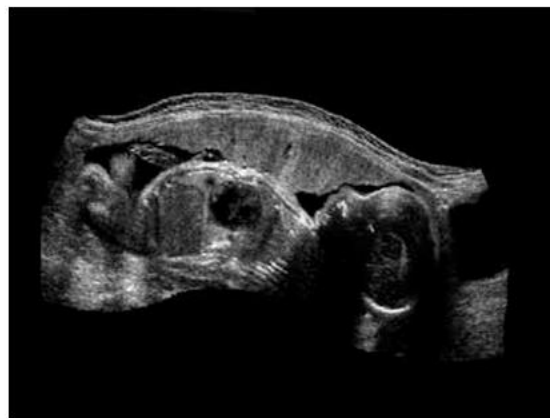


FIGURE 1.18 Images of the Crab Pulsar (in the center of each image) covering the electromagnetic spectrum. (Courtesy of NASA.)

FIGURE 1.19

Cross-sectional image of a seismic model. The arrow points to a hydrocarbon (oil and/or gas) trap. (Courtesy of Dr. Curtis Ober, Sandia National Laboratories.)

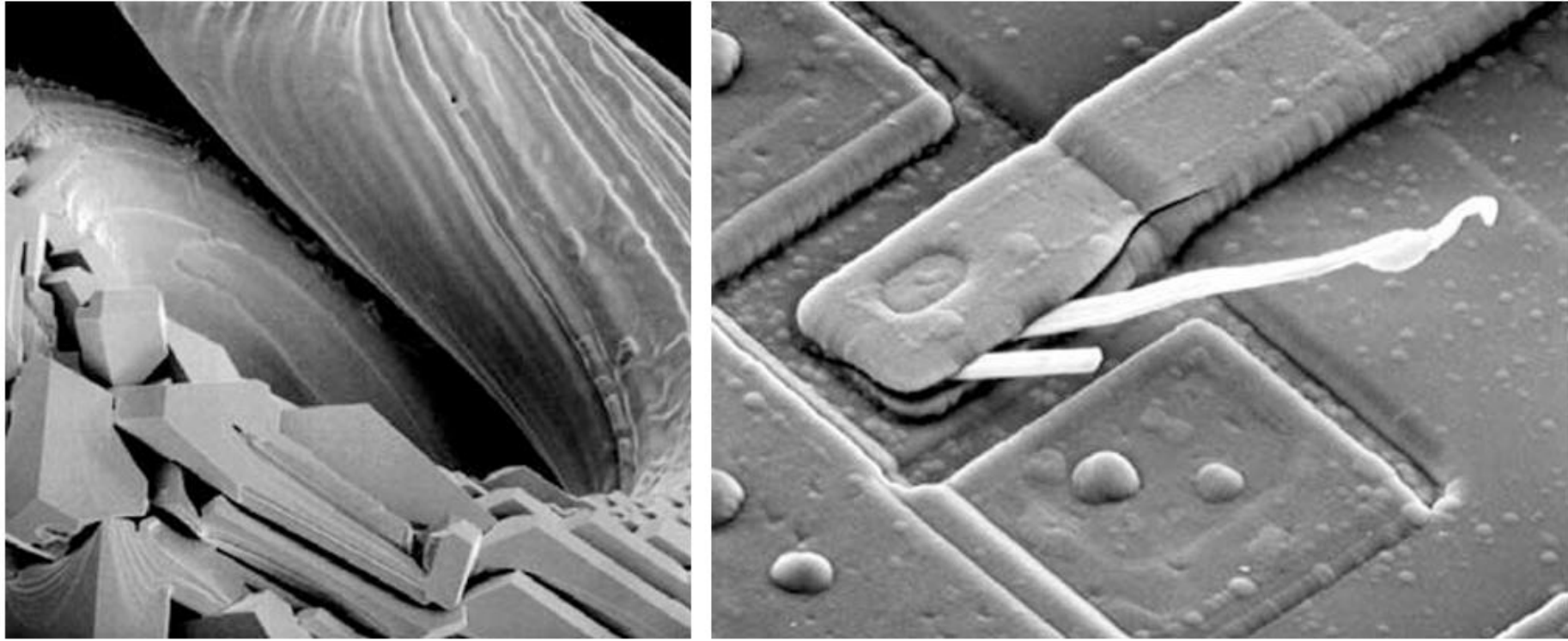




a	b
c	d

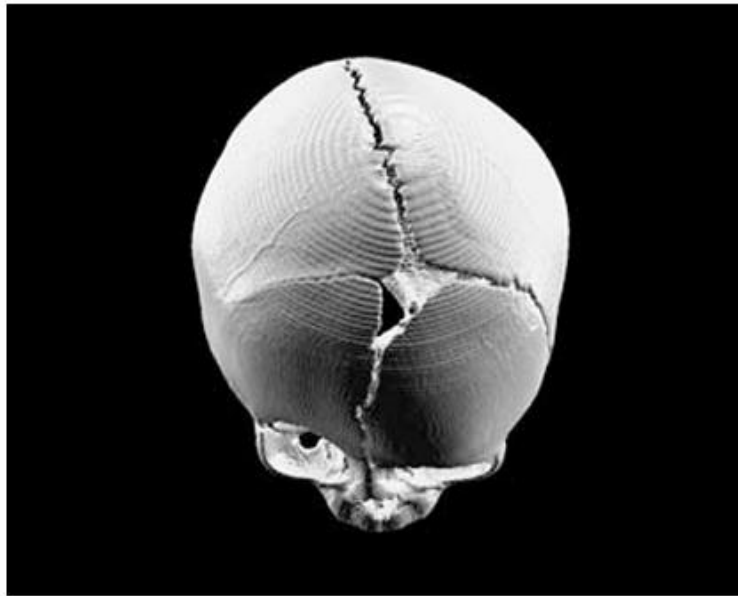
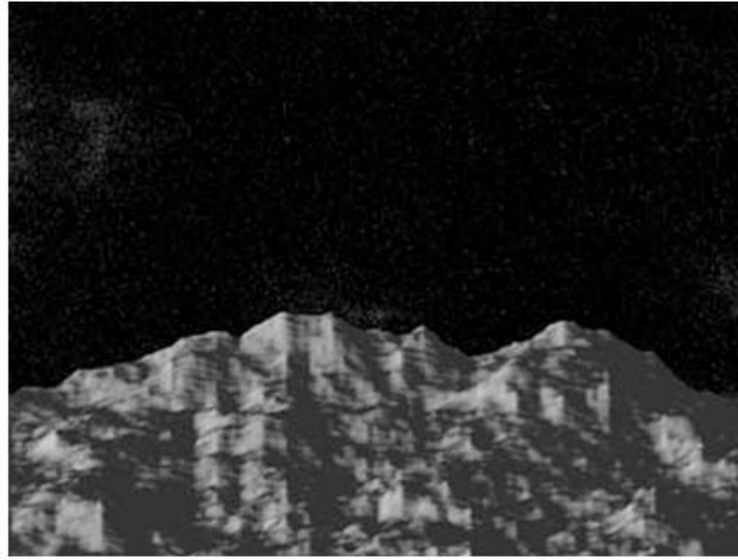
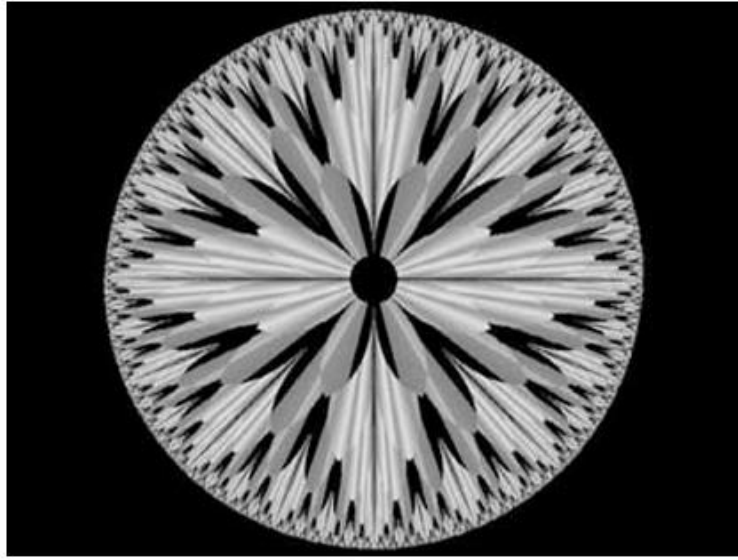
FIGURE 1.20

Examples of ultrasound imaging. (a) Baby. (b) Another view of baby. (c) Thyroids. (d) Muscle layers showing lesion. (Courtesy of Siemens Medical Systems, Inc., Ultrasound Group.)



a b

FIGURE 1.21 (a) 250 $\times$ SEM image of a tungsten filament following thermal failure (note the shattered pieces on the lower left). (b) 2500 $\times$ SEM image of damaged integrated circuit. The white fibers are oxides resulting from thermal destruction. (Figure (a) courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene; (b) courtesy of Dr. J. M. Hudak, McMaster University, Hamilton, Ontario, Canada.)



a	b
c	d

FIGURE 1.22
 (a) and (b) Fractal
 images. (c) and
 (d) Images
 generated from
 3-D computer
 models of the
 objects shown.
 (Figures (a) and
 (b) courtesy of
 Ms. Melissa
 D. Binde,
 Swarthmore
 College; (c) and
 (d) courtesy of
 NASA.)

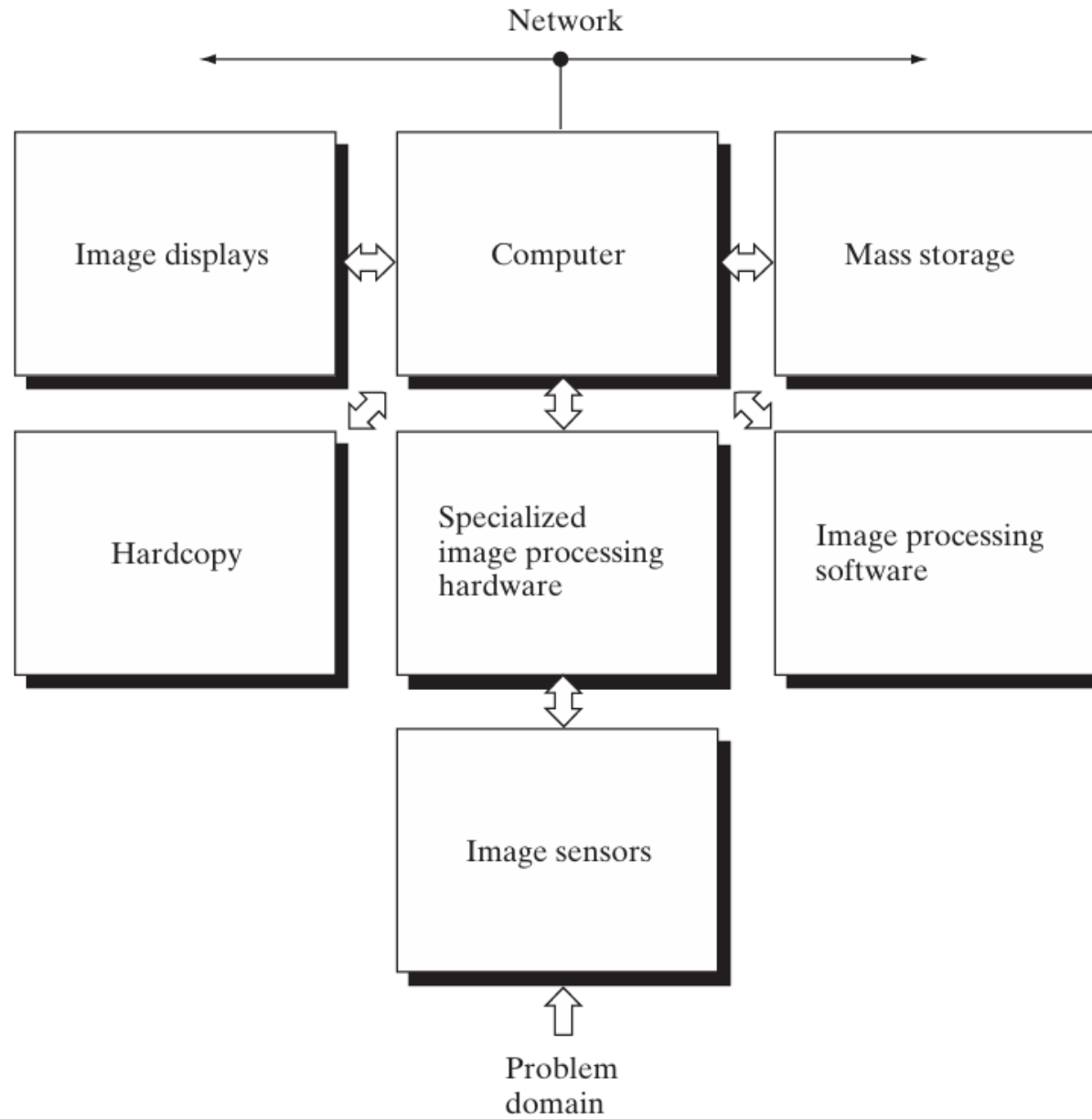


FIGURE 1.24
Components of a
general-purpose
image processing
system.

1. Infrared (Thermal) Images



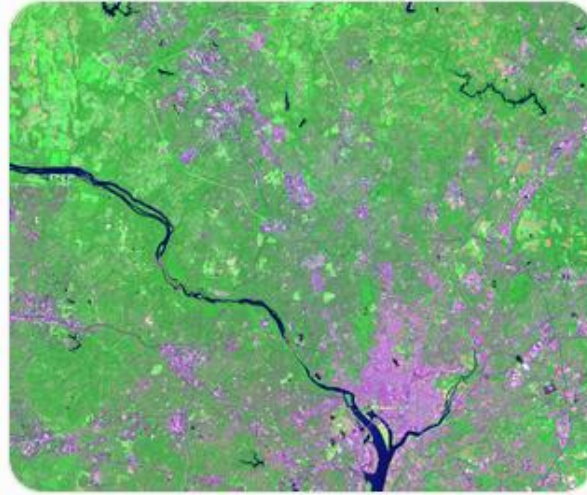
Applications

- Fever screening
- Fire detection
- Building heat-loss inspection
- Electrical fault detection
- Night vision
- Search and rescue

Characteristics

- Usually grayscale or false-color
- Represents **temperature**, not visible color
- Wavelength: **0.75 μm –14 μm** (near IR to thermal IR)

2. Multispectral Images



Applications

Agriculture

Forest monitoring

Water resources

Land-use mapping

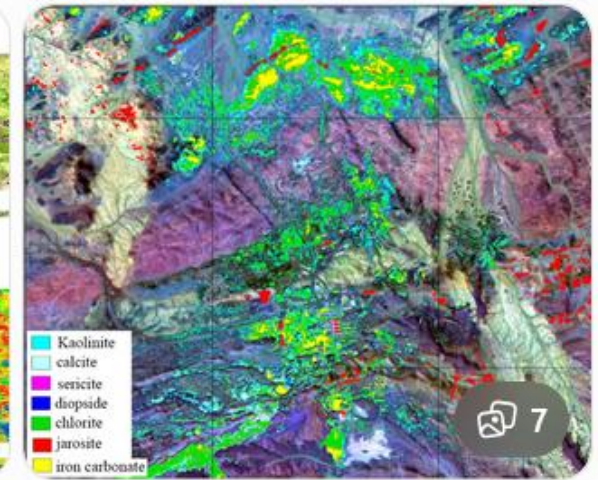
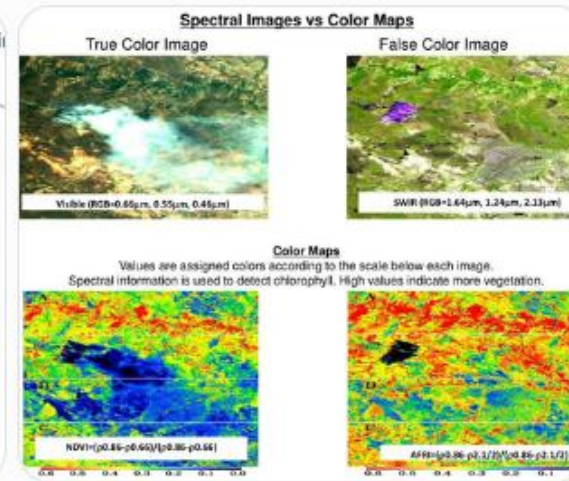
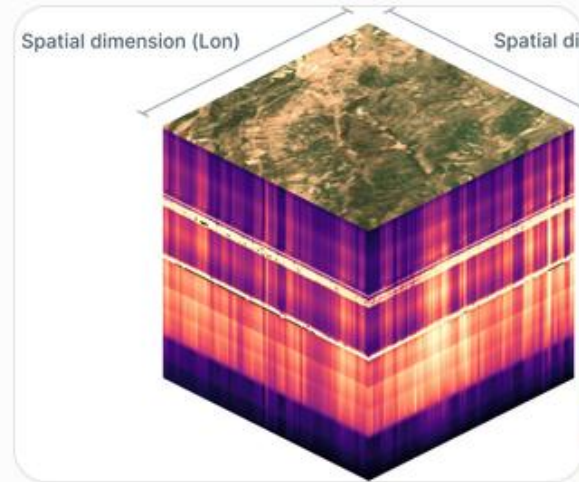
Disaster management

Characteristics

Typically 4–13 spectral bands

Example bands: *Blue, Green, Red, Near Infrared (NIR) Shortwave Infrared (SWIR)*

3. Hyperspectral Images



Applications

- Mineral exploration
- Precision agriculture
- Food quality inspection
- Medical imaging
- Environmental monitoring

Characteristics

- **100–300+ very narrow spectral bands**
- Each pixel contains a **spectral signature**, enabling identification of different materials

Comparison of spectral resolution

Image Type	Typical Bands	Measures	Major Applications
RGB Image	3 (R,G,B)	Visible colors	Photography, Computer Vision
Infrared	1–3	Temperature / Heat	Thermal imaging, Night vision
Multispectral	4–13	Reflectance in selected wavelength bands	Agriculture, Remote sensing
Hyperspectral	100–300+	Detailed spectral signature	Material identification, Mineral mapping, Crop health

Spatial Resolution



Classroom Explanation

Suppose you photograph the **same scene** with different image resolutions.

Resolution	Image Size	Quality
High	4000 × 3000 (12 MP)	Sharp, fine details visible
Medium	1920 × 1080 (2 MP)	Good quality for screens
Low	640 × 480	Details start disappearing
Very Low	320 × 240	Pixelated, blurry

Processing and Remote Sensing.



Interpretation

Spatial Resolution	Example Sensor	What You Can Observe
0.3–0.5 m	WorldView / GeoEye	Individual vehicles, trees, buildings, roads
10 m	Sentinel-2	Crop fields, rivers, forests, urban blocks
30 m	Landsat	Land-use classes, lakes, large roads, vegetation

Quantization in DIP...*Contd.....*Radiometric Resolution

Quantization	Gray Levels	MATLAB Data Type	Example
1-bit	2	logical	Black & White
2-bit	4	uint8	4 gray levels
4-bit	16	uint8	Simple grayscale
8-bit	256	uint8	Standard grayscale image
12-bit	4096	uint16	Medical imaging (CT, X-ray)
16-bit	65536	uint16	MRI, scientific cameras
24-bit	16.7 million colors	RGB (uint8)	Color images

Quantization In DIP

Image Type	Typical MATLAB Data Type	Why?
Binary (1-bit)	logical	Only black and white
Grayscale Photo	uint8	256 gray levels are sufficient
RGB Photo	uint8	8 bits per color channel
Medical Images (CT, MRI)	uint16	High intensity precision
Thermal Infrared	uint16	Accurate temperature measurements
Multispectral	uint16 or single	Reflectance values need greater precision
Hyperspectral	single, double, or uint16	Hundreds of bands; scientific computations
Deep Learning Input	single	Faster GPU computation and lower memory usage
Image Processing Algorithms	double	High numerical precision for mathematical operations

Quantization in DIP...*Contd*

MATLAB Data Type	Bits	Value Range	Typical Images	Memory/Pixel
logical	1	0 or 1	Binary image	1 bit
uint8	8	0–255	Grayscale, RGB photos	1 Byte
uint16	16	0–65535	CT, MRI, scientific cameras	2 Bytes
int16	16	–32768 to 32767	Scientific measurements	2 Bytes
single	32	Decimal numbers	Image processing algorithms	4 Bytes
double	64	Decimal numbers	Mathematical computations	8 Bytes

Question?